A Knowledge Graph for Colon Adenocarcinoma

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BIOT 6900 · Module 7 Assignment

My disease is colon adenocarcinoma (COAD), carried forward from Module 2. The graph runs on Neo4j AuraDB Free and has 54 nodes and 61 relationships: 18 genes (15 COAD, 3 for the related disease), 2 diseases, 5 phenotypes, 9 pathways, 13 drugs and 7 variants.

1. Schema and evidence

I used the six node types from the worked example, because each one is a step in the question the graph has to answer: which drugs act on genes linked to my disease, and which drugs for a similar disease might be worth a look? I left out the optional patient layer, because my only cohort (Module 6) has a simulated phenotype, and including it would make the graph look like it held clinical data when it does not.

I picked the 15 COAD genes so that every kind of evidence from Part I is in the graph, rather than taking the top 15 of one list:

- Multi-omics (Module 2), 4 genes: CA2, DES, CR2, MYH11. My strongest RNA-and-protein genes (CA2 RNA −5.20, protein −2.16). All four are lower in tumour.
- Somatic drivers (Module 2 CPTAC), 2 genes: KRAS (35 of 106 tumours) and BRAF (18 of 106). Both are significant on the mutation layer; none of my multi-omics genes were.
- Approved COAD drug targets, 5 genes: EGFR, VEGFA, TYMS, TOP1, PTGS2. My own genes are all down in tumour, so without these the 2-hop query would only find drugs for down-regulated genes, which is the wrong direction.
- Germline GWAS loci (Module 6 PRS), 4 genes: MYC (rs6983267, $p = 2 \times 10^{-122}$), SMAD7 (rs4939827, 3×10^{-30}), TCF7L2 (rs11196172, 7×10^{-20}), BMP4 (rs4444235, 8×10^{-10}).

Each gene also carries my Module 3 tractability prior, $\max(p_{sm}, p_{ab})$, refit with every graph gene held out of training. Every edge carries a type, a source and a confidence level, and gene-to-disease edges carry a p-value where one exists.

Relationship	n	source values	confidence	Other property
Gene –ASSOCIATED_WITH→ Disease	18	multi-omics (TCGA/CPTAC), somatic mutation (CPTAC), pharmacology, GWAS Catalog	high / medium / low	pvalue (GWAS genes)
Disease –HAS_PHENOTYPE→ Phenotype	8	HPO	low	—
Gene –PARTICIPATES_IN→ Pathway	15	Reactome	low	—
Drug –TARGETS→ Gene	13	ChEMBL	high	drug status on the node
Variant –OCCURS_IN→ Gene	7	GWAS Catalog (Module 6), CPTAC somatic (Module 2)	high (somatic) / low (GWAS)	—

Table 1. Relationship types and the evidence on each edge. Confidence levels come from the ranking in Section 4.

The source matters most on the 18 gene-to-disease edges, because those hold four very different kinds of claim. “This protein drops 4-fold in tumour” is not the same evidence as “a COAD drug targets this gene”, and the source label is the only thing that tells them apart in a query.

The assignment asks for Module 5 variants, but my Module 5 gene was CYP2C19, which has no link to colon cancer. I used variants about my own disease instead: four COAD GWAS lead SNPs from Module 6, and three somatic hotspots from the Module 2 CPTAC tumours (KRAS G12D 11 of 106, KRAS G12C 2 of 106, BRAF V600E 13 of 106).

For identifiers, every gene enters as an HGNC symbol from my Module 3 proteome table, with its Ensembl ID attached. Neo4j has a uniqueness constraint on name and I load with MERGE, so re-running cannot create a duplicate gene. The one weak point is GWAS SNPs, where the Catalog's mapped gene is sometimes not the gene I attached (rs6983267 maps to CASC8/CCAT2, but MYC is the gene it is thought to regulate). I attached the literature candidate gene and recorded the Catalog's mapping in data/targets_coad.csv.

2. Top targets

Rank	Gene	PageRank	Evidence type	Druggability
1	KRAS	0.0377	somatic mutation	0.50
2=	BRAF	0.0231	somatic mutation	0.69
2=	EGFR	0.0231	pharmacology	0.71
2=	TYMS	0.0231	pharmacology	0.09
5=	CA2, MYC, SMAD7, TCF7L2, BMP4, TOP1, PTGS2, VEGFA	0.0158	mixed	0.02–0.77
last	DES, MYH11, CR2	0.0086	multi-omics	0.00–0.64

Table 2. Candidate biomarker ranking. PageRank fused with the gene evidence.

The top three match the biology. KRAS, BRAF and EGFR are central to how COAD is treated: RAS and BRAF status decide whether a patient gets anti-EGFR antibodies, and the KRAS G12C and BRAF V600E inhibitors are approved for these tumours.

But they are on top because of my schema, not the biology. A gene only gains PageRank from edges pointing into it, and the only edges pointing into genes come from drugs and variants. KRAS is first because two drugs and two variants point at it. Any gene with neither sits at the floor value of 0.0086. So the ranking measures how many drugs and variants I attached, and I chose those.

Two results stand out. TYMS ties for second despite weak COAD evidence (RNA +0.85, no protein measurement), a druggability prior of 0.09 and no GWAS signal. It ranks there only because fluorouracil and capecitabine point at it. TYMS is a chemotherapy target needed by any dividing cell, so this is the clearest case of the graph rewarding pharmacology over disease biology. In the other direction, MYC has the strongest genetic evidence in the graph ($p = 2 \times 10^{-122}$, over a hundred orders of magnitude stronger than BMP4) but only ranks mid-table, because PageRank never looks at the p-value. My table shows PageRank and evidence side by side instead of merging them into one number.

Community detection put DES, MYH11 and CR2 in one cluster with the COAD node, the “Smooth muscle contraction” and “Complement cascade” pathways and two symptoms. I read this cluster as tissue composition, not tumour mechanism. DES and MYH11 are smooth-muscle proteins and CR2 is a B-cell receptor. Normal colon samples contain muscle and lymphoid tissue, while tumour samples are mostly epithelium, so these genes look “lost in tumour” partly because the normal samples contain more muscle. Read this way, the cluster is a warning about my own Module 2 ranking.

3. A hypothesis from a path

The 2-hop query returned the approved COAD drugs acting on genes in my graph: cetuximab and panitumumab through EGFR, bevacizumab through VEGFA, fluorouracil and capecitabine through TYMS, irinotecan through TOP1, sotorasib through KRAS and encorafenib through BRAF. The shared-symptom query then returned two repurposing candidates from ulcerative colitis. One full path:

> **Ustekinumab –TARGETS [ChEMBL]→ IL12B –ASSOCIATED_WITH [GWAS Catalog, $p = 1 \times 10^{-21}$]→ Ulcerative colitis –HAS_PHENOTYPE [HPO]→ Iron deficiency anaemia ←HAS_PHENOTYPE [HPO]– Colon adenocarcinoma**

The second candidate was tofacitinib through JAK2, by the same route. The path is weak: the two diseases are only linked by three general symptoms (rectal bleeding, abdominal pain, anaemia), so taken literally it would

suggest a UC drug for any disease that makes people bleed. I treat the path as a pointer and take the hypothesis from the gene at the end of it.

IL12B codes for p40, the subunit shared by IL-12 and IL-23. IL-23 drives gut inflammation, and UC itself raises colorectal cancer risk. My hypothesis is that blocking IL-23 reduces the growth of colorectal tumours that start on an inflammatory background. One complication the graph cannot see: ustekinumab blocks p40, so it also blocks IL-12, which helps the immune system attack tumours. A drug blocking only the IL-23-specific p19 subunit, such as mirikizumab, would be a cleaner test.

To test it, I would first compare an IL-23/IL-17 signature (IL23A, IL12B, IL23R, IL17A) between tumour and normal in TCGA-COAD, data I already have. If that holds up, I would run the AOM/DSS colitis-associated cancer mouse model with three arms (anti-p40, anti-p19, control antibody) and count tumours.

4. An edge is evidence, not truth

My edge types, weakest first. These rankings are what the confidence values in Table 1 record.

Pharmacology gene-to-disease edges (5). The weakest, and circular. I linked EGFR, VEGFA, TYMS, TOP1 and PTGS2 to COAD because a COAD drug targets them, so when the 2-hop query reports that COAD drugs act on COAD genes, part of that answer is something I put in. These edges also drive the ranking in Section 2. With Open Targets association scores instead, I would expect TYMS and TOP1 to drop out.

Phenotype edges (8). I typed these in by hand from the HPO term list, and all five COAD phenotypes are very general. My whole repurposing result passes through them. With proper HPO annotations, or more specific phenotypes such as chronic mucosal inflammation, UC would probably still connect but the symptom-only paths would disappear.

GWAS variant-to-gene edges (4). OCCURS_IN is simply wrong for at least two. rs6983267 sits in an enhancer the Catalog maps to CASC8/CCAT2, and rs4444235 maps near BMP4, not in it. Colocalisation with colon eQTLs would show which gene each SNP really acts on, and the edge should be called REGULATES.

Pathway edges (15). Labelled Reactome, but they are my own simplified groupings, and four of my nine pathways hold a single gene. Full Reactome membership would merge several of my communities, so the neat one-pathway-one-cluster picture in Section 2 is partly an artefact of how sparse I kept this layer.

Multi-omics edges (4). Real and highly significant, but Section 2 shows they may be measuring tissue composition, and the RNA and protein layers come from different patients. Single-cell or deconvolved expression would separate real epithelial loss from sample composition, and I would expect DES and MYH11 to lose their edges.

Drug-to-target and somatic-variant edges (20). The strongest; mechanisms are established and I counted the hotspots from the CPTAC table myself. The catch is direction: acetazolamide inhibits CA2, but CA2 is lost in tumour, so the edge is true and still not useful for therapy.

One problem runs through the whole graph: every edge counts the same. PageRank treats a pathway edge I typed in by hand exactly like a GWAS hit at 10^{-122}. A weighted PageRank using $-\log_{10} p$ for genetic edges would fix this, and I would expect MYC and SMAD7 to rise and TYMS to fall.

I used an AI assistant (Claude) for editing this report.